Project: Worked examples
Title: Helical springs
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1. Scope and method

A cylindrical helical spring of round wire, cold-coiled, with a linear characteristic: compression (rate, working points, solid length by end type, stress at the working points and at solid, buckling, Goodman fatigue, surge frequency), extension (the same with the initial tension and the hook stresses) or torsion (rate with the legs, curvature-corrected bending stress, wound diameter and body length).

WHAT IT CALCULATES	RELATION
Index and rate	$C = D/d$; $k = G \cdot d^4/(8 \cdot D^3 \cdot n)$ (compression, extension); $k_\varphi = E \cdot d^4/(64 \cdot D \cdot n_a)$, $n_a = n + (l_1 + l_2)/(3\pi \cdot D)$ (torsion)
Spring stress	$\tau_k = k_c \cdot 8 \cdot F \cdot D/(\pi \cdot d^3)$, $k_c = (C + 0.5)/(C - 0.75)$ (Bergsträsser) or $(4C - 1)/(4C - 4) + 0.615/C$ (Wahl); torsion $\sigma = q \cdot 32 \cdot M/(\pi \cdot d^3)$ with $q = (C + 0.07)/(C - 0.75)$ (EN 13906-3, the default) or $K_i = (4C^2 - C - 1)/(4C \cdot (C - 1))$ (Shigley)
Solid length (compression)	$n_t = n + e_n$; EN 13906-1 form $L_s = d_{max} \cdot (n_t + e_s)$ with $(e_n, e_s) = (0, 1)$ plain, $(1, 0)$ plain and ground, $(2, 1.5)$ squared (closed, unground), $(2, 0)$ squared and ground, and the minimum coil clearance $S_a = (0.0015 \cdot D^2/d + 0.1 \cdot d) \cdot n$ checked at the working length ($L_2 \geq L_s + S_a$); Shigley form $L_s = d \cdot (n_t + e_s)$ with $e_s = 1$ for squared; pitch $p = (L_o - a \cdot d)/(n + b)$
Extension spring hook stresses	$F = F_o + k \cdot s$; hook bending $\sigma_A = F \cdot [K_A \cdot 16 \cdot D/(\pi \cdot d^3) + 4/(\pi \cdot d^2)]$, $K_A = (4C_1^2 - C_1 - 1)/(4C_1 \cdot (C_1 - 1))$, $C_1 = 2r_1/d$; hook torsion $\tau_B = K_B \cdot 8 \cdot F \cdot D/(\pi \cdot d^3)$, $K_B = (4C_2 - 1)/(4C_2 - 4)$, $C_2 = 2r_2/d$
Buckling (compression)	$\lambda_{eff} = \nu \cdot L_o/D$; $y_{cr} = L_o \cdot C_1' \cdot [1 - \sqrt{(1 - C_2'/\lambda_{eff}^2)}]$, $C_1' = E/(2(E - G))$, $C_2' = 2\pi^2(E - G)/(2G + E)$; stable at any deflection when $\lambda_{eff}^2 < C_2'$
Fatigue (Goodman, as M4)	$1/n_f = a/S_e + m/S_u$ with $a = (x_2 - x_1)/2$, $m = (x_2 + x_1)/2$ of the stresses at the two working points; $S_u = f_{su} \cdot R_m$ in shear, R_m in bending; R_m typed or $R_m = A/d\text{^}m$ from a W3 material (A in MPa, d in mm)
Surge frequency	$f_n = (d/(2\pi \cdot n \cdot D^2)) \cdot \sqrt{(G/(2\rho))}$; warning when $f_n/f_{op} < 13$
Torsion body	$D_w = D \cdot n/(n + \varphi/2\pi)$; $L_w = d \cdot (n + 1 + \varphi/2\pi)$

Basis: Rate, stress correction, solid length, buckling, fatigue, surge[1](Calculation of compression springs); Rate, initial tension, hook stress[2](Calculation of extension springs); Rate, bending stress with curvature correction[3](Calculation of torsion springs); Helical springs: end geometry, buckling, critical frequency, extension and torsion springs[4](10th ed., Ch. 10).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Kind of spring	Compression		spec: Plunger assembly PLG-210 §4 rev B	any
corr	Stress correction factor k_c for the shear stress	Bergsträsser		spec: Company spring standard SPR-01 (EN 13906-1 correction factor)	any
d	Wire diameter	3	mm	drawing: PLG-210-04 rev B	0.1 … 20 mm
D	Mean coil diameter	24	mm	drawing: PLG-210-04 rev B	0.5 … 400 mm
n	Active coils (torsion: coils of the body)	9		drawing: PLG-210-04 rev B	2 … 200
ends	End type (compression)	Squared and ground		drawing: PLG-210-04 rev B	any
$solid_{method}$	Solid length and clearance (compression): EN 13906-1 uses d_{max}, 1.5 d more for squared unground ends, and checks the minimum coil clearance at L_2; Shigley uses the nominal d	EN 13906-1 (d_max, coil clearance)		spec: Company spring standard SPR-01 (EN 13906-1 solid length and minimum coil clearance); no wire tolerance entered, so d_max is the nominal 3 mm	any
L0	Free length (compression)	70	mm	drawing: PLG-210-04 rev B	1 … 1000 mm
work	Working points given as	Loads		spec: Plunger assembly PLG-210 §4 rev B	any
F_1	Force at working point 1 (loads)	60	N	spec: Plunger assembly PLG-210 §4 rev B	0 … 20000 N
F_2	Force at working point 2 (loads)	180	N	spec: Plunger assembly PLG-210 §4 rev B	0 … 20000 N
G	Shear modulus	79.3	GPa	assumption: Generic spring steel shear modulus; take it from the W3 material of the wire certificate.	20 … 100 GPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
E	Young's modulus	206	GPa	assumption: Generic spring steel Young's modulus; take it from the W3 material of the wire certificate.	50 … 250 GPa
ρ	Density	7850	kg/m³	assumption: Generic steel density.	1000 … 20000 kg/m³
rm_{source}	Where $R_m(d)$ comes from: typed (no default: the wire certificate or your wire standard) or the curve R_m = A/d^m of a W3 material	Typed R_m(d)		assumption: R_m(d) typed for this example; a W3 material with the curve R_m = A/d^m can be applied instead.	any
Rm	Tensile strength of the wire at this diameter $R_m(d)$, typed (no default)	1650	MPa	assumption: Example value for a 3 mm wire. Enter R_m(d) from your own copy of the wire standard or the wire certificate (record spring.wire-strength); the standard's table is not shipped.	300 … 3500 MPa
f_{tau}	Permissible shear stress as a fraction of R_m (generic textbook default; replace with the licensed EN 13906 value)	0.45		assumption: Static permissible shear stress fraction of R_m for the example; take it from your design reference (record spring.permissible).	0.2 … 0.8
f_{su}	Ultimate shear strength as a fraction of R_m (Goodman, shear; generic textbook default, replace with the licensed EN 13906 value)	0.67		assumption: Ultimate shear strength fraction of R_m used by the Goodman relation (textbook steel value); confirm in your reference.	0.4 … 0.9
v_{seat}	End-support coefficient v (0.5 both ends fixed, 0.7 one fixed one pinned, 1 both pinned, 2 one free)	0.5		spec: Plunger assembly PLG-210 §4: both end plates flat and parallel, no rotation	0.5 … 2

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
S_e	Endurance strength for the stress kind (shear; bending in torsion) at N_c cycles, alternating (no default: from the wire maker or your fatigue reference)	450	MPa	assumption: Example alternating shear endurance strength at 10^7 cycles; take it from the wire maker or your fatigue reference (record spring.endurance).	50 … 2000 MPa
N_c	Required life the endurance strength belongs to	10000000		spec: Plunger assembly PLG-210 §4: duty life	10000 … 10000000000
$n_{f,req}$	Required Goodman safety factor	1.2		spec: Company spring standard SPR-01	1 … 5
f_{op}	Operating (excitation) frequency	8	Hz	spec: Plunger assembly PLG-210 §4: maximum cycle rate	0 … 1000 Hz

3. Assumptions

A1. Linear characteristic: the rate is constant between the working points; no progressive or conical spring, no variable pitch. (Cold-coiled round wire up to 20 mm.)
A2. Active coils n are as entered; end coils are accounted for only in the solid length (and the pitch) by the end type; the hook and leg flexibility is in the rate only for the torsion legs. (Shigley 10th ed. Ch. 10, geometry of the ends.)
A3. Solid length: the EN 13906-1 form (default) uses the maximum wire diameter d_{max}, adds the 1.5 d of an unground squared end and checks the minimum coil clearance S_a at the working length L_2; plain ends have no EN row and keep the geometric form. The Shigley form (nominal d, no clearance) is selectable. (EN 13906-1 relations written in own notation, to be checked against the owner's copy; Shigley 10th ed. Table 10-1.)
A4. Torsion bending stress: the EN 13906-3 factor q = (C + 0.07)/(C − 0.75) is the default; Shigley's K_i = (4C² − C − 1)/(4C(C − 1)) is selectable and about 1 % lower at C = 8. (EN 13906-3 relation written in own notation, to be checked against the owner's copy.)
A5. R_m(d) and the endurance strength are the user's inputs from the user's own copy of the wire standard and design reference, with no default. The permissible stress fractions f_τ, f_σ and f_{su} do default (0.45, 0.75, 0.67), to generic textbook fractions that the user replaces with the licensed EN 13906 value. R_m(d) can instead be taken from the curve R_m = A/d^m of a W3 material (revision-pinned; A in MPa for d in mm, the diameter range of the fit is the user's to respect). No wire-strength table and no permissible-stress diagram is shipped. (No wire-strength table or permissible-stress diagram is shipped.)
A6. The stress correction factor k_c is applied to the working and the solid stress alike (conservative for the static check). (Choose 'None (uncorrected)' for the uncorrected static stress; the fatigue stresses (Goodman) always carry the curvature factor, the Bergsträsser one when 'None' is chosen.)
A7. Extension: hook fatigue and the hook shape beyond its two radii are not assessed; the initial tension is a force the user enters. (Shigley 10th ed. Ch. 10, extension springs.)
A8. Surge is the first natural frequency of a spring with both ends fixed; a spring with one end free has half of it. (Shigley 10th ed. Ch. 10, critical frequency.)
A9. G (Shear modulus) is taken as 79.3 GPa. (Generic spring steel shear modulus; take it from the W3 material of the wire certificate.)
A10. E (Young's modulus) is taken as 206 GPa. (Generic spring steel Young's modulus; take it from the W3

material of the wire certificate.)
A11. ρ (Density) is taken as 7850 kg/m³. (Generic steel density.)
A12. rm_{source} (Where $R_m(d)$ comes from: typed (no default: the wire certificate or your wire standard) or the curve R_m = A/d^m of a W3 material) is taken as Typed $R_m(d)$. ($R_m(d)$ typed for this example; a W3 material with the curve R_m = A/d^m can be applied instead.)
A13. Rm (Tensile strength of the wire at this diameter $R_m(d)$, typed (no default)) is taken as 1650 MPa. (Example value for a 3 mm wire. Enter $R_m(d)$ from your own copy of the wire standard or the wire certificate (record spring.wire-strength); the standard's table is not shipped.)
A14. f_{tau} (Permissible shear stress as a fraction of R_m (generic textbook default; replace with the licensed EN 13906 value)) is taken as 0.45. (Static permissible shear stress fraction of R_m for the example; take it from your design reference (record spring.permissible).)
A15. f_{su} (Ultimate shear strength as a fraction of R_m (Goodman, shear; generic textbook default, replace with the licensed EN 13906 value)) is taken as 0.67. (Ultimate shear strength fraction of R_m used by the Goodman relation (textbook steel value); confirm in your reference.)
A16. S_e (Endurance strength for the stress kind (shear; bending in torsion) at N_c cycles, alternating (no default: from the wire maker or your fatigue reference)) is taken as 450 MPa. (Example alternating shear endurance strength at 10^7 cycles; take it from the wire maker or your fatigue reference (record spring.endurance).)

4. Working

[7] C = D / d [1](/ EN 13906-2 / EN 13906-3)
= 24 mm / 3 mm
= 8
Spring index
[15] k_c = (C + 0.5) / (C - 0.75)
= (8 + 0.5) / (8 - 0.75)
= 1.1724
Shear stress correction factor (Bergsträsser)
[16] $k_{c,fat}$ = k_c
= 1.1724
= 1.1724
Shear stress correction factor for the fatigue stresses (as the static stress)
[17] τ_{perm} = f_{tau} * Rm
= 0.45 × 1650 MPa
= 742.5 MPa
Permissible shear stress (fraction of R_m, user input)
[20] k = G * d^4 / (8 * D^3 * n) [1]
= 79.3 GPa × (3 mm)^4 / (8 × (24 mm)^3 × 9)
= 6.4535 N/mm
Spring rate
[21] n_t = n + 2
= 9 + 2
= 11
Total coils (derived: end coils added to the active coils)
[22] d_{max} = d
= 3 mm
= 3 mm
Maximum wire diameter (none entered: the nominal d)
[23] L_s = d_{max} * (n_t + 0.0) [1]
= 3 mm × (11 + 0.0)
= 33 mm
Solid length L_c (EN 13906-1 form: maximum wire diameter)

[24] $S_a = (0.0015 * D^2 / d + 0.1 * d) * n$ [1]
$= (0.0015 \times (24\ mm)^2 / 3\ mm + 0.1 \times 3\ mm) \times 9$
$= 5.292\ mm$
Minimum sum of the coil clearances S_a
[25] $L_n = L_s + S_a$ [1]
$= 33\ mm + 5.292\ mm$
$= 38.292\ mm$
Minimum spring length $L_n = L_c + S_a$
[26] $p = (L0 - 2 * d) / (n + 0)$
$= (70\ mm - 2 \times 3\ mm) / (9 + 0)$
$= 7.1111\ mm$
Pitch of the free spring (derived)
[27] $s_c = L0 - L_s$
$= 70\ mm - 33\ mm$
$= 37\ mm$
Travel to solid
[28] $F_c = k * s_c$
$= 6.4535\ N/mm \times 37\ mm$
$= 238.78\ N$
Force at solid length
[32] $s_1 = F_1 / k$
$= 60\ N / 6.4535\ N/mm$
$= 9.2974\ mm$
Deflection at working point 1
[33] $s_2 = F_2 / k$
$= 180\ N / 6.4535\ N/mm$
$= 27.892\ mm$
Deflection at working point 2
[34] $\tau_1 = k_c * 8 * F_1 * D / (pi * d^3)$ [1]
$= 1.1724 \times 8 \times 60\ N \times 24\ mm / (\pi \times (3\ mm)^3)$
$= 159.23\ MPa$
Shear stress at working point 1
[35] $\tau_2 = k_c * 8 * F_2 * D / (pi * d^3)$ [1]
$= 1.1724 \times 8 \times 180\ N \times 24\ mm / (\pi \times (3\ mm)^3)$
$= 477.68\ MPa$
Shear stress at working point 2
[36] $\tau_c = k_c * 8 * F_c * D / (pi * d^3)$ [1]
$= 1.1724 \times 8 \times 238.78\ N \times 24\ mm / (\pi \times (3\ mm)^3)$
$= 633.67\ MPa$
Shear stress at solid length
[37] $u_2 = \tau_2 / \tau_{perm}$
$= 477.68\ MPa / 742.5\ MPa$
$= 0.64335$
Utilisation at working point 2
[38] $u_c = \tau_c / \tau_{perm}$
$= 633.67\ MPa / 742.5\ MPa$
$= 0.85343$
Utilisation at solid length
[39] $L_1 = L0 - s_1$
$= 70\ mm - 9.2974\ mm$
$= 60.703\ mm$
Spring length at working point 1

[40] $L_2 = L0 - s_2$
= 70 mm − 27.892 mm
= 42.108 mm
Spring length at working point 2
[41] $u_s = s_2 / s_c$
= 27.892 mm / 37 mm
= 0.75384
Working deflection / travel to solid
[42] $s_n = L0 - L_n$
= 70 mm − 38.292 mm
= 31.708 mm
Travel to the minimum length L_n
[43] $u_n = s_2 / s_n$
= 27.892 mm / 31.708 mm
= 0.87965
Working deflection / travel to the minimum length (coil clearance at L_2)
[44] $\lambda_{eff} = v \cdot L_e / D$ [1]
= 0.5 × 70 / 24
= 1.4583
Effective slenderness
[45] C1p = E/(2(E − G)) [4](10th ed. Ch. 10, buckling)
= 206 / (2 × (206 − 79.3))
= 0.81294
Buckling constant C_1'
[46] $C2p = 2\pi^2(E - G)/(2G + E)$ [4](10th ed. Ch. 10, buckling)
= $2\pi^2$ × (206 − 79.3) / (2 × 79.3 + 206)
= 6.8595
Buckling constant C_2'
[47] u_b = stable at any deflection ($\lambda_{eff}^2 < C_2'$)
= 2.1267 < 6.8595
= 0
Deflection at point 2 against the buckling deflection (none: stable)
[48] f_n = d / (2 * pi * n * D^2) * sqrt(G / (2 * ρ)) [1](/ EN 13906-2)
= 3 mm / (2 × π × 9 × (24 mm)^2) × sqrt(79.3 GPa / (2 × 7850 kg/m³))
= 207 Hz
Surge (natural) frequency, both ends fixed
[50] $r_f = f_n / f_{op}$
= 207 Hz / 8 Hz
= 25.875
Surge frequency / operating frequency
[55] $\tau_a = k_{c,fat} / k_c * (\tau_2 - \tau_1) / 2$
= 1.1724 / 1.1724 × (477.68 MPa − 159.23 MPa) / 2
= 159.23 MPa
Alternating shear stress
[56] $\tau_m = k_{c,fat} / k_c * (\tau_2 + \tau_1) / 2$
= 1.1724 / 1.1724 × (477.68 MPa + 159.23 MPa) / 2
= 318.46 MPa
Mean shear stress
[57] $S_{su} = f_{su} * Rm$
= 0.67 × 1650 MPa
= 1105.5 MPa
Ultimate shear strength (fraction of R_m, user input)

[58] $u_f = \tau_a/S_e + \tau_m/S_{su}$ [4](10th ed. Ch. 6))

= 159.23 / 450 + 318.46 / 1105.5

= 0.64191

Goodman utilisation ($1/n_f$)

[59] $n_f = 1/u_f$

= 1 / 0.64191

= 1.5579

Goodman safety factor

[60] $u_{fat} = n_{f,req} * u_f$

= 1.2 × 0.64191

= 0.77029

Fatigue utilisation against the required factor

5. Results

SYMBOL	VALUE	DESCRIPTION
k	6.5 N/mm	Rate
F_1	60.0 N	Force at point 1
F_2	180.0 N	Force at point 2
L_s	33.0 mm	Solid length
f_n	207.00 Hz	Surge frequency
C	8	Spring index
Rm	1650.0 MPa	Wire tensile strength $R_m(d)$
k_c	1.1724	Stress correction factor
n_t	11	Total coils
p	7.1 mm	Pitch
s_1	9.3 mm	Deflection at point 1
s_2	27.9 mm	Deflection at point 2
L_1	60.7 mm	Length at point 1
L_2	42.1 mm	Length at point 2
s_c	37.0 mm	Travel to solid
F_c	238.8 N	Force at solid
τ_1	159.2 MPa	Stress at point 1
τ_2	477.7 MPa	Stress at point 2
τ_c	633.7 MPa	Stress at solid
d_{max}	3.0 mm	Maximum wire diameter
S_a	5.3 mm	Minimum sum of coil clearances
L_n	38.3 mm	Minimum length L_n
s_n	31.7 mm	Travel to L_n
y_{cr}	"— (stable at any deflection)"	Buckling deflection
buckling	"PASS"	Buckling

SYMBOL	VALUE	DESCRIPTION
r_f	25.875	Surge ratio f_n/f_{op}
n_f	1.56	Goodman safety factor
u_2	0.64	Utilisation at point 2
u_c	0.85	Utilisation at solid
u_s	0.75	Working deflection / travel to solid
u_n	0.88	Working deflection / travel to L_n (governing)
u_b	0.00	Deflection at point 2 / buckling deflection
u_f	0.64	Goodman utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Minimum coil clearance at L_2: working deflection within the travel to $L_n = L_s + S_a$	u_n = 0.88	$u_n \le 1$	0.88	[1]
PASS	Static shear stress at solid length	u_c = 0.85	$u_c \le 1$	0.85	[1]
PASS	Goodman fatigue against the required safety factor	u_{fat} = 0.77	$u_{fat} \le 1$	0.77	[5](M4)
PASS	Working deflection within the travel to solid	u_s = 0.75	$u_s \le 1$	0.75	[1]

7. Applicability

WARNING records: tabulated values not taken from a reference record are G, E, ρ (Material properties); Rm (Wire tensile strength $R_m(d)$); f_{tau}, f_{su} (Permissible and ultimate stress fractions); v_{seat} (End-support coefficient); S_e (Endurance strength). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

```
Template mech.spring@1.3.2 Content hash 0ba43838cef719e9 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. EN 13906-1.
2. EN 13906-2.
3. EN 13906-3.
4. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
5. Goodman, as.